Design and Implementation of a Intelligent Virtual Agent for Kabré Supercomputer Infrastructure Support

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Abstract. abstract

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1 Problem Definition

High-Performance Computing (HPC) environments typically host highly heterogeneous user communities composed of students, researchers, and professionals from diverse disciplines. This diversity naturally leads to a wide range of technical expertise, resulting in queries of varying complexity—from basic questions about system usage and command-line operations to advanced issues involving software installation, resource optimization, and memory management.

In Costa Rica, the Kabré supercomputer, operated by the National High Technology Center (CeNAT), represents a critical infrastructure for scientific research. The system currently supports approximately 300 active users, generating over 100,000 computing hours annually. This community is primarily composed of students, faculty members, and researchers from public universities across the country, spanning multiple domains of knowledge. Despite its importance, user support poses a significant operational challenge, as it is managed by a small team consisting of a single infrastructure engineer and one technical assistant.

At present, user support is handled through a ticket-based system, where users submit queries that may include problem descriptions, code snippets, and supporting files. While functional, this approach presents several limitations. First, it introduces friction in the user experience, as users must navigate the platform, authenticate, and formally structure their requests before receiving assistance. Second, resolving issues often requires multiple back-and-forth interactions to clarify context, which increases response and resolution times.

Furthermore, the system becomes overloaded due to both the volume of requests and the recurrence of similar queries. Many of these correspond to frequently asked questions that require repetitive responses, leading to inefficient

use of the support team's time. At the same time, more specialized queries—such as those arising in fields like bioinformatics or computational sciences—require in-depth, domain-specific analysis. This combination of routine and complex queries contributes to system inefficiency and the accumulation of unresolved tickets.

Although a knowledge base exists within the current system, it is outdated, limited in scope, and lacks interactivity. Users cannot perform targeted queries or obtain immediate answers, forcing them to rely entirely on the ticketing system. As a result, response times can exceed 24 hours, which is particularly problematic in scenarios requiring rapid troubleshooting or near real-time assistance.

Given these challenges, there is a clear need to leverage modern technological solutions to improve user support in HPC environments. In particular, the development of an intelligent virtual agent emerges as a promising approach to automate the handling of user queries, especially those that are repetitive in nature. Such a system could provide real-time assistance, reduce the operational burden on infrastructure staff, and enhance the overall user experience. However, in its initial stages, the agent should not operate with full autonomy; instead, it should work in conjunction with human experts, determining which queries can be effectively resolved automatically and which require escalation to the support team.

2 Literature Review

The evolution of High-Performance Computing (HPC) has undergone a fundamental shift, moving from purely hardware-centric challenges toward the complex management of user interaction and resource allocation in increasingly heterogeneous environments. As evidenced by the operational constraints of the Kabré supercomputer, the gap between an expanding, diverse user base and limited administrative personnel creates a critical bottleneck that traditional, static support systems are unable to resolve. To address this, contemporary literature explores the transition toward autonomous and semi-autonomous management frameworks. These approaches seek to replace rigid administrative structures with dynamic, intelligent layers capable of handling the high-dimensional demands of modern scientific computing while democratizing access for non-expert researchers.

Within this architectural shift, the work of Bychkov, Oparin, Feoktistov, Bogdanova, and Sidorov (2017) provides a foundational theoretical basis for rethinking HPC support through intelligent agents. They argue that traditional supercomputing administration is no longer sufficient to cope with the inherent dynamism of current environments. Instead, they propose a service-oriented system of intelligent agents that replaces fixed protocols with a more flexible, fluid, and adaptive architecture. Their core contribution lies in the application of aggregate models and dynamic resource allocation, demonstrating that large-scale scientific problems, including stability analysis and SAT-related computations, can be addressed with near-perfect operational efficiency [9]. By implementing a

structure capable of real-time simulation and failure monitoring, this approach seeks to close the gap created by the lack of high-level tools for multi-agent orchestration, reframing HPC administration as a responsive framework that continuously adapts to the evolving demands of diverse scientific workloads [9].

Complementing the structural management of these environments, recent research emphasizes the necessity for more sophisticated interaction layers. Investigations by Guo, Zhang, Soong, He, and Xie (2020) propose the integration of context encoders in Text-to-Speech (TTS) systems to achieve more natural, human-like speech generation without relying on manual annotation. This synergy is vital for an HPC support agent, as it combines the operational efficiency of a multi-agent architecture with the communicative depth required to assist a diverse scientific community effectively [2]. Furthermore, Ashish, Jaiswal, Vhaduri, Nerella, and Jha (2025) demonstrate that human-centric skills, such as intercultural collaboration, can be embedded into computing curricula through AI-based virtual mentors like Empa. Utilizing lightweight language models such as $LLaMA$ 3, they enable the creation of "cultural scaffolding" systems that help researchers navigate the interpersonal complexities of global HPC projects, although they identify persistent gaps in formal curricular integration and the long-term evaluation of these soft skills [6].

The convergence of Artificial Intelligence and HPC also demands renewed resource management strategies, as traditional methods often fail to optimize performance and energy consumption simultaneously. Narantuya, Shin, Park, and Kim (2022) demonstrate that a decentralized Multi-Agent Deep Reinforcement Learning ($mDRL$) model significantly outperforms conventional policies by considering future rewards rather than just current states. This approach allows for a 20% reduction in task completion times and a 40% decrease in inefficient energy consumption [16]. Despite these advancements, the study identifies critical gaps regarding economic accessibility for smaller research institutes and the limitations of centralized clusters in meeting low-latency requirements. This highlights an open need for integration with Edge Computing to process data near its source, ensuring that intelligent management layers can optimize resource usage within critical scientific infrastructures like the Kabré environment [16].

Building on the need for more efficient support systems, the research conducted by Jain, Gupta, and Neha (2024) underscores that traditional ticket management systems are no longer sufficient to handle the complexity of contemporary IT environments. A central finding of their work is that semantic similarity alone is inadequate for effective incident clustering; instead, it is crucial to integrate spatial and temporal factors, such as network topology and event timing. By adopting this multidimensional approach, their system achieves a clustering fidelity of 96% and effectively reduces manual effort by 30% [18]. However, the authors identify a significant lack of adaptability in current commercial platforms and a major hurdle in how organizations transition from legacy systems to AI-driven frameworks without compromising specialized institutional knowledge or facing internal cultural resistance [18].

The challenge of fragmented information is further addressed by Bondapalli, Zheng, Ajayi, Keceli, and colleagues (2025), who observe that HPC documentation is often scattered across multiple formats and repositories. They argue that general-purpose language models frequently fail in this domain due to outdated knowledge and technical hallucinations. Their core finding is that a Retrieval-Augmented Generation (RAG) framework, enhanced by multimodal processing that preserves the structure of code and tables, allows their system to outperform direct AI queries with a relevance rating of 4.7/5.0 [20]. Despite these results, they note that the primary bottleneck for accuracy is the inclusion of noisy or irrelevant context, identifying the need for advanced retrieval techniques such as hybrid search and knowledge graphs to move beyond static documentation [20].

Further advancing the paradigm of autonomy, Rosendo, DeWitt, Souza, and their collaborators (2025) argue that traditional scientific workflow systems are excessively static and rely on error-prone manual coordination. To solve this, they propose a modular architecture based on AI agents that integrate natural language interfaces with programmable facility APIs. This orchestration enables adaptive and explainable decision-making, accelerating scientific discovery through near real-time resource coordination [4]. However, the research highlights that integrating workflows across different facilities is still labor-intensive due to voids in policy and security protocols. Specifically, current security policies in HPC centers, designed for batch processing, are not suited for interactive or real-time streaming workflows, representing an unmet obstacle for autonomous experimentation [4].

Broadening the focus toward the user experience, Biggs and Dalton (2024) identify a critical knowledge gap among researchers, noting that expertise in one specific system does not guarantee proficiency in another. They demonstrate that implementing an AI assistant through RAG allows for precise, site-specific technical assistance based on updated internal documentation, avoiding the high costs of constant model retraining [5]. Nevertheless, they identify the difficulty of processing noise in historical support tickets—such as personal information or irrelevant conversations—as an open challenge for converting records into useful knowledge sources. This is echoed by Diederich et al. (2020), who provide a comprehensive analysis of the conversational agent (CA) landscape, arguing that many systems still fail to foster positive experiences because they overlook individual user characteristics such as culture or personality. They propose an agenda prioritizing longitudinal field studies to verify whether social cues in agents can lead to long-term sustainability in scientific infrastructures [22].

Expanding the framework of intelligent infrastructure management, Oparin, Bogdanova, and Pashinin (2020) base their research on developing microservice compositions within the $HPCSOMAS$ framework. They demonstrate that integrating static hierarchical control with dynamic, decentralized $P2P$ agent networks provides exceptional flexibility and scalability for complex problems like binary dynamic systems [25]. Despite these breakthroughs, the inherent complexity of heterogeneous distributed environments remains a barrier for specialists who are not expert programmers. The authors highlight a persistent lack

of high-level tools for the transparent integration of cloud resources with high-performance environments, leaving open the question of how to adapt agent behavior models to allow for even more sophisticated hybrid functioning [25].

Addressing the operational inefficiencies caused by user behavior, Zaborovsky, Utkin, Muliukha, and Lukashin (2023) find that supercomputer users drastically overestimate task execution times in over 89% of cases. This leads to inefficient queues and low resource density. To resolve this, they integrate a multi-agent scheduler with machine learning models based on survival analysis, utilizing Random Survival Forests (RSF) to predict probability functions rather than simple fixed values. While this improves estimation accuracy, the study emphasizes the need for Explainable AI (XAI) to provide technical support that helps users understand why a task failed and how to adjust parameters [14]. Furthermore, they propose a transition toward Natural Language Processing (NLP) interfaces, allowing scientists to engage with supercomputers through meaningful dialogues rather than complex configuration scripts.

The human element is central to the research of Monroy, Becerra, Bellas, and Duro (2006, 2008), who observe that HPC centers often prioritize technical metrics like wait times while overlooking the true objective of maximizing user satisfaction. They identify that users frequently overestimate required resources due to a lack of technical knowledge or fear of task abortion. By implementing the $EVOPROC$ virtual interface, which utilizes neural networks and evolutionary algorithms, they demonstrate that it is possible to predict actual resource usage far more accurately than original user requests [1,**?**]. Their work addresses the technological void of tools that translate imprecise human requests into optimal operational requirements, transforming HPC center administration into an adaptive process that balances technical efficiency with a positive perception of service quality [1].

Finally, early pioneering efforts by Faigle, Fox, Furmanski, Niemiec, and Simoni (1993) established that virtual environments (VR) and HPC are complementary technologies that must be integrated to enhance large-scale simulations. They argue that advanced "Grand Challenge" simulations require dynamic VR front-ends for visualization, and they demonstrate the necessity of decoupling the update rates of HPC compute engines from those of VR peripherals to prevent simulation sickness [7]. Despite these early advancements, they identified a lack of flexible and standardized connectivity models and a significant void in hardware-independent user interface protocols. These challenges remain relevant today as the scientific community strives for "Televirtuality" and collaborative discovery, underlining a persistent gap between the VR and HPC research communities that contemporary agent-based support systems aim to bridge [7]. Collectively, these twenty sources illustrate a broader transition in HPC research: from purely computational optimization toward hybrid systems that integrate intelligent resource management, human-centered design, and adaptive, autonomous interaction layers.

3 Research Questions

Based on the challenges identified in the Kabré HPC support system, this study explores the design, implementation, and impact of an Intelligent Virtual Agent (IVA). The research is guided by the following main questions:

1. To what extent can an Intelligent Virtual Agent improve the efficiency of technical support in the Kabré supercomputer environment, particularly in terms of response time, resolution time, and overall system performance?
2. To what extent can the introduction of an IVA reduce the workload of the technical support team by handling repetitive and frequently asked queries?

In addition to these main questions, the study also considers several complementary aspects. These include the role of multimodal features such as blendshapes for lip-sync and whether they provide meaningful value in user interaction or remain largely irrelevant in an HPC support context. It also examines the impact of the IVA on user satisfaction, the types of data that can be used to train the underlying language model, and how data quality influences system performance. Furthermore, the study explores how autonomy can be designed so that the agent can decide which queries to resolve automatically and which to escalate to human experts. Security considerations are also addressed, particularly in scenarios where the IVA is granted elevated permissions, requiring safeguards to prevent risks to the HPC infrastructure. Finally, the study evaluates which types of queries can be reliably handled by the IVA, based on validation with domain experts, as well as the system's ability to communicate in a clear and empathetic manner across a diverse user base.

4 Objectives of the Agent

The Intelligent Virtual Agent (IVA) is designed to act as a first-line support system for the Kabré High-Performance Computing (HPC) infrastructure. Its main objectives are:

1. To provide real-time assistance to users by resolving common technical issues related to system usage, job execution, software installation, and resource management.
2. To reduce the workload of the human support team by handling repetitive and frequently asked questions, while ensuring that complex cases are escalated when necessary.
3. To improve the overall user experience within the HPC environment by offering clear, adaptive, and accessible guidance tailored to different user profiles.

5 Preliminary Technical Architecture

The architecture of the Intelligent Virtual Agent (IVA) is designed as an integrated system combining large language models, voice services, and an interactive interface to support the technical assistance process within the Kabré HPC

environment. The design is motivated by the heterogeneity of the user community, which implies that interactions cannot follow a single uniform behavior. Instead, the agent is expected to adapt its responses according to predefined user clusters such as social sciences, natural sciences, or computational disciplines, an approach consistent with recent work on LLM-based support systems in HPC environments [21] and findings in conversational agent research highlighting the importance of user characteristics and adaptive interaction design in socio-technical systems [22]. This adaptation affects both the level of technical detail and the explanatory style used in responses.

At the core of the system, the Gemma 4 E4B model is used, deployed within the Kabré infrastructure. This model is selected due to its ability to operate efficiently in high-performance computing environments and its strong performance in natural language understanding and generation tasks. These characteristics make it suitable for handling a wide range of technical support queries, from basic system usage to more complex issues such as job execution, software installation, and resource management.

Interaction with the system is based exclusively on textual input from the user. Users communicate with the agent only through text, without any speech input capability. However, the system incorporates VoxCPM2 as a text-to-speech (TTS) engine for the agent's output. In this configuration, the language model generates textual responses that are subsequently converted into speech, enabling a multimodal output experience while maintaining a simple and consistent text-only input interface. The use of speech synthesis in this context is motivated by advances in conversational neural TTS systems, which show that incorporating conversational context and semantic awareness significantly improves prosody naturalness and perceived realism in voice agents [23]. Such approaches demonstrate that high-quality conversational TTS benefits from modeling dialogue context and utterance-level dependencies, leading to more natural and contextually appropriate speech generation.

The user interface is developed using Unity, enabling a real-time interactive environment for communication with the agent. This interface serves as the main point of interaction between users and the support system, complementing the traditional ticket-based workflow. The use of an avatar-based representation of the agent is intended to make the interaction more intuitive and accessible, particularly for less technical users. This design choice is consistent with research on interactive agent-based avatar systems, where embodied virtual agents have been shown to improve usability and engagement by combining visual representation with structured behavioral models [24].

Additionally, the system may incorporate facial animation technologies such as the NVIDIA LipSync model to improve synchronization between generated speech and visual representation. This component is considered optional and is intended to enhance the naturalness of the interaction experience.

The system is designed to run primarily within the Kabré infrastructure, leveraging its computational resources for both the language model and backend services. The IVA will be accessible exclusively through the official Kabré

website, operating as an online service integrated into the institutional portal, where users can interact with the agent directly from their web browser.

Overall, this architecture is intended to address the main limitations identified in the current support system, including ticket overload, user heterogeneity, and long response times. Rather than replacing human support, the system is designed to complement it by automating repetitive queries while escalating complex cases to human experts when necessary.

6 GitHub Repository

The GitHub repository contains the project structure and documentation of the intelligent virtual agent, including the design proposal and supporting materials for its implementation in HPC support for the Kabré environment. It also includes the initial project organization, with space reserved for the future integration of code components.

https://github.com/DylanBC09/Intelligent-Virtual-Agent-for-Kabr-Supercomputer-Infrastructure-Support-/blob/main/README.md

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7 GitHub Repository